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Project Thesis

submitted in partial fulfillment of the
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A comparision of various deep learning based detection systems for multi-object tracking

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Abstract

[Put your abstract here.]

General structure of an abstract, write 1 to 2 sentences per section

1. A general statement introducing the broad research area of the particular topic being investigated.
2. An explanation of the specific problem (difficulty, obstacle, challenge) to be solved.
3. A review of existing or standard solutions to this problem and their limitations.
4. An outline of the proposed new solution.
5. A summary of how the solution was evaluated and what the outcomes of the evaluation were.

You may find the following resources useful:

- `https://www.grammarly.com/blog/write-an-abstract/`
- `https://www.editage.com/insights/manuscript-structure-how-to-convey-your-most-im`
- More useful links: `https://hps.vi4io.org/teaching/ressources/start`

Declaration on the use of ChatGPT and comparable tools in the context of examinations

In this work I have used ChatGPT or another AI as follows:

- ☑ Not at all
- ☐ During brainstorming
- ☐ When creating the outline
- ☐ To write individual passages, altogether to the extent of 0% of the entire text
- ☐ For the development of software source texts
- ☐ For optimizing or restructuring software source texts
- ☐ For proofreading or optimizing
- ☐ Further, namely: -

I hereby declare that I have stated all uses completely.
Missing or incorrect information will be considered as an attempt to cheat.

Contents

List of Tables **iv**

List of Figures **v**

List of Listings **vi**

List of Abbreviations **vii**

1 Introduction **1**

2 Related work **1**
2.1 Overview of modern detection systems 1
2.2 Integration of detection systems in tracking systems 2

3 Experimental setting **3**
3.1 Datasets 3
3.1.1 MOT15 Dataset collection 3
3.1.2 MOT20 Dataset collection 4
3.2 Detection systems 4
3.3 Multi-object-tracking systems 4

4 Benchmark results **5**
4.1 Metrics 5
4.2 Analysis 5

5 Discussion **6**

A Additional Materials **A1**

List of Tables

List of Figures

1 Every detection system in time 2
2 3

List of Listings

List of Abbreviations

HPC High-Performance Computing

1 Introduction

2 Related work

2.1 Overview of modern detection systems

Edozie et. al **edozie** provides a review across the past developments of object detection techniques in the past 26 years. Besides describing the evolution of deep learning based object detection systems they also compare them to the classical non-deep-learning based object detection methods and algorithms.

They provide a wide overview of datasets, methods and research that was important in the past 26 years. Beside providing classical object detection methods they also present an overview of modern deep learning based object detector which should be summarized here to give an impression of the state of the art.

In their work they present different ways to categorize object detection systems. One way object detection systems are separated in One-Stage and Two-Stage object detectors. Where the two-stage detector first generates proposals of the detection and then encodes them to make class predictions. Otherwise the first-stage detector doesn't consider region proposals, they simply rely on the locations with the most objects and either classify them as an object or background. Through this simplification One-Stage detectors are generally faster than two-stage detectors.

This categorization approach leaves out other architectures like transformer-based models like DETR <**empty citation**> and Multi-path CNNs like PAN <**empty citation**> Instead they summarize the different detection architectures by the categories of tradional detectors which include non deep learning methods, conventional deep learning methods that only inlcude two-stage detection systems, HybridBriding which provides a category for Multi-path detection systems and advanced deep learning which is a category for one-stage detectors and transformer based models. To present some methods figures **??** an overview of different detectors and their category in consideration of their year of development provided by **edozie**.

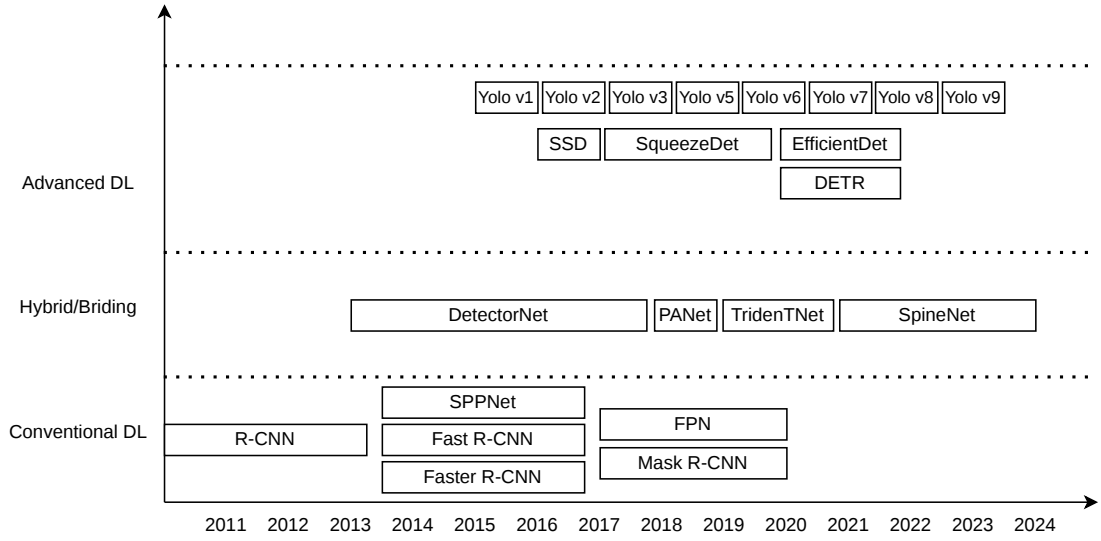

Figure 1: The graphic summarizes the recent years developments in detection systems: R-CNN <**empty citation**> SPPNet <**empty citation**> Fast R-CNN <**empty citation**> Faster R-CNN <**empty citation**> FPN <**empty citation**> Mask R-CNN <**empty citation**> DetectorNet <**empty citation**> PANet <**empty citation**> TridenNet <**empty citation**> SpineNet <**empty citation**> SSD <**empty citation**> SqueezeDet <**empty citation**> EfficientDet <**empty citation**> DETR <**empty citation**>

Neha et al. **1** provides an overview of detection systems using deep learning and classical approaches. They provide a summary of datasets, metrics and methods relevant in this domain. Further they provide a list of strengths and weaknesses of the different deep-learning based detection systems and also provide benchmark results on various detection systems, e.g.: R-CNN, Yolo etc.

2.2 Integration of detection systems in tracking systems

3 Experimental setting

3.1 Datasets

The choice of dataset is important to correctly reflect the actual performance of different permutations of detection and tracking systems. Especially newer datasets that include night-time scenes, non-urban environments, or highly occluded images can provide useful insights into performance improvements over time. Furthermore, this section highlights the evolution of various dataset collections and challenges.

3.1.1 MOT15 Dataset collection

Different permutations of tracking and detection systems were evaluated on the MOT15 dataset **MOTChallenge2015**. The MOT15 challenge is a dataset framework that contains 5503 training frames and 5783 testing frames, divided into a total of 22 sequences. The dataset includes both 3D and 2D sequences; however, for the current benchmark, only the 2D sequences are considered.

Furthermore, the dataset provides a diverse set of scenes and categorizes the data according to camera movement (static or moving), viewpoint height, and weather conditions. The ground truth annotations are primarily generated through manual human labeling, supported by annotation tools such as the VATIC annotation tool. Some sequences originate from existing datasets, such as the ETH <**empty citation**> dataset and the KITTI <**empty citation**> dataset, which provide their own annotated data. In Figure 2, exemplary ground truth annotations are presented.

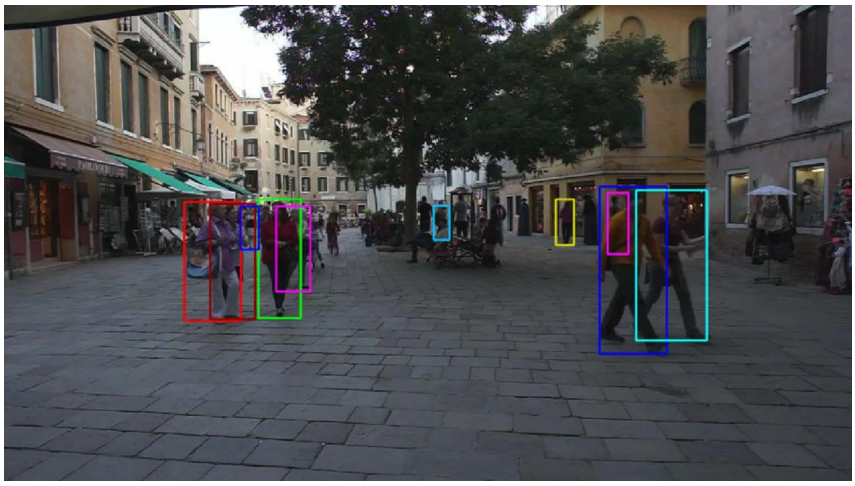

Figure 2: Example scene from the Venice-2 sequence, which is included in the MOT15 dataset. The frames show the point of view (POV) of a static camera in the inner city of Venice. The bounding boxes represent the ground truth annotations provided by the MOT15 challenge.

This figure shows one of the targets used in the benchmark. Most sequences, such as KITTI-17, TUD-Crossing, and ETH-Crossing, depict urban environments. The dataset mainly consists of scenes in which multiple pedestrians are visible. As a result, occlusion is common due to overlapping pedestrians. Many sequences originate from datasets such as ETH-Bahnhof, ETH-Sunnyday, and ETH-Pedcross2.

As previously mentioned, the dataset is divided into two parts: the training set and the testing set. While the training and testing sets contain data from similar environments, they are recorded at different times. For example, the KITTI-17 sequence is from 2017 and is located in the training set, while the KITTI-19 sequence is from 2019 and is located in the testing set. This structure makes it possible to train and evaluate detection and tracking models under similar but temporally distinct conditions. Since this report focuses on pre-trained models, the evaluation is independent of whether the training or testing dataset is used.

3.1.2 MOT20 Dataset collection

Since the MOT15 dataset primarily includes daytime scenes, an additional dataset from a newer MOTChallenge benchmark is considered. The MOT20 dataset **dendorfer2020mot20benchmar** includes scenes recorded at night and in more crowded environments. This provides additional challenges for detection and tracking systems and allows evaluation under more difficult real-world conditions.

3.2 Detection systems

3.3 Multi-object-tracking systems

4 Benchmark results

4.1 Metrics

4.2 Analysis

5 Discussion

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A Additional Materials